Hood Wheel Ratio Test

Which hood wheel speed makes the ball spin the least?

Team: _____ Date: _____ Names: _____

The big idea

Our shooter squeezes the ball between **two wheels**: a **big flywheel** (about 4 inches) in front and a **small hood wheel** (about 2 inches) in the back. If the two wheel surfaces move at *different* speeds, the ball starts to **spin**. A spinning ball can curve or bounce in a way we do not want.

The small wheel has to turn **faster** to keep up with the big wheel. The number that tells us how much faster is the **ratio**:

> **Ratio = hood wheel speed ÷ flywheel speed**
>
> Example: flywheel 2000 RPM and ratio 2.00 → hood wheel = 2000 × 2.00 = **4000 RPM**

Our guess: a ratio near **2.00** gives the least spin. **Our job:** test it, and find out if the best ratio stays the same when the flywheel goes faster or slower.

SAFETY — everybody must follow these

- ☐ **Eye protection on** for everyone near the shooter.
- ☐ **Nobody stands in front** of the shooter or in the path of the ball.
- ☐ **Hands, hair, sleeves and lanyards stay away** from the wheels.
- ☐ **Wheels keep spinning after you stop them.** Wait until they are completely stopped before you reach in.
- ☐ Anyone can yell **"STOP!"** The Driver presses **X** right away.
- ☐ **Only a coach clears a jam.**

Jobs (one person can do two jobs if you are a small group)

Driver	Holds controller 1. Turns the dials, starts and stops the shooter, flips the gate. Says "Shooting!" before each shot.
Feeder	Holds controller 2. Runs the intake (Right Bumper) to load balls.
Video	Films every shot in **slow motion**. Says the shot number out loud.
Recorder	Writes everything in the data table. Reads the screen.
Spotter	Watches where the ball lands and watches for safety. Calls "STOP!" if needed.

Set up (do this once)

- ☐ Fresh battery. Battery volts: _____
- ☐ **Coach check:** the encoder check and the gate positions are done. Coach initials: _____
- ☐ Put a tape mark on the floor. Shooting distance: _____ inches. **Use the same mark for every shot.**
- ☐ Draw a **thick marker stripe** all the way around the ball. This is how we see the spin.
- ☐ Set the phone to **slow motion** on a steady stand where it can see the ball leave the shooter.
- ☐ On the Driver Station choose **"Shooter Ratio - Alpha"**. Press INIT. The gate should say **LOADING**. Press START.

Controls

Controller 1 (Driver)

B / A	Flywheel RPM up / down
D-pad Right / Left	Ratio up / down
Y	Shooter ON
X	**Shooter OFF** (STOP)
Left Trigger	Flip the gate
Back (Share)	Reset the "lowest RPM"

Controller 2 (Feeder)

Right Bumper	Intake: pull balls in
Left Bumper	Reverse the intake

Reading the screen

At Speed? true = OK to shoot.
CAPPED = the hood wheel cannot go that fast. **Skip that row.**
Lowest = the slowest each wheel went while the ball went through.

For every row of the data table

1. **Driver:** set the flywheel RPM (B / A) and the ratio (D-pad) to match the row. Check the hood wheel RPM on the screen matches the table. If it says CAPPED, skip the row.
2. **Driver:** press **Y**. Wait for **At Speed? true**.
3. **Feeder:** load a ball. **Driver:** say "Shooting!" and flip the gate with the **Left Trigger**. Flip it back after the ball leaves.
4. **Video** says "Shot 1". **Recorder** fills in the spin code and the landing letter (see below).
5. Wait for **At Speed? true** again. Repeat until you have **5 shots**.
6. **Recorder:** write the **Lowest** flywheel RPM and hood wheel RPM from the screen. Then the Driver presses **Back**.
7. Go to the next row. (Changing a dial also resets the "lowest RPM".)

How to write down a shot

Watch the slow-motion video. Follow the marker stripe from the moment the ball leaves until it lands.

Top box: spin code (a letter + a number)

B	**Backspin**: the top of the ball turns **back toward the robot**
N	**No spin**: the stripe hardly moves (use **N0**)
T	**Topspin**: the top of the ball turns **forward toward the target**

Number: 0 = none 1 = a little (stripe goes around less than 2 times) 2 = medium (2 to 6 times) 3 = a lot (more than 6). Examples: **B2**, **N0**, **T1**.

Bottom box: where it landed

S	**Short**: landed before the target
O	**On** the target
L	**Long**: landed past the target

> The wheels keep spinning after you press X. **Never reach in until they stop.**

Part 1 — Data Table (first distance)

Distance: _____ Battery volts: _____ Recorder: _____

Each shot box: **top** = spin code (like B2) | **bottom** = landing (S, O or L). Hood RPM = flywheel RPM × ratio.

Flywheel RPM	Ratio	Hood RPM	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Lowest flywheel RPM	Lowest hood RPM
1500	**1.80**	2700							
1500	**2.00**	3000							
1500	**2.20**	3300							
1500	**2.40**	3600							
2000	**1.80**	3600							
2000	**2.00**	4000							
2000	**2.20**	4400							
2000	**2.40**	4800							
2250	**1.80**	4050							
2250	**2.00**	4500							
2250	**2.20**	4950							
2250	**2.40**	5400							
2700	**2.00**	5400							

Bonus row (last row): ask your coach first. The hood wheel is at its 5400 RPM limit. If the screen says CAPPED, do not record it. Every other row stays at or under 5400, so none of them should say CAPPED.

Part 2 — Change the distance, then the results

Part 2: second distance

Move the shooter to a **new distance** and mark it. Ask your coach which 3 flywheel speeds reach the target from there. Use your **best ratio from Part 1**, and also the ratio **0.20 lower** and **0.20 higher** than it. Keep every hood wheel RPM at **5400 or less**.

New distance: _____ Best ratio from Part 1: _____ Battery volts: _____

Flywheel RPM	Ratio	Hood RPM	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Lowest flywheel RPM	Lowest hood RPM

Results

Flywheel RPM	Ratio with the **least spin** (distance 1)	Spin code there	Ratio with the least spin (distance 2)
1500			
2000			
2250			

Think about it

1. Was the best ratio about the **same** at every speed? Circle one: **YES** **ALMOST** **NO**

2. Did the **lowest RPM** of the hood wheel or the flywheel drop more when the ball went through? Which wheel? _____

3. What do you think is the reason?

_____

4. Did the best ratio change when you changed the **distance**?

_____